



TED UNIVERSITY

Faculty of Engineering

Department of Computer Engineering

CMPE 491 Senior Project

ARIS

Analysis Report

Fall 2025

Taha Akdemir | ID: 12577008730

Ecem Nur Bilgi | ID: 14830754170

Şevkiye Sıla Kahya | ID: 14620089482

Elif Zeynep Elverişli | ID: 13942218426

TABLE OF CONTENTS

1. Introduction	4
1.1 General Overview	4
1.2 Constraints	4
1.3 Professional and Ethical Issues	6
2. Current System.....	6
3. Proposed System.....	7
3.1 Overview	7
3.2 Functional Requirements	7
3.2.1 Visual & Audio Input Capture Requirement.....	8
3.2.2 Command Recognition Requirement	9
3.2.3 Communication Module Requirement	9
3.2.4 User Interface Requirement.....	10
3.2.5 System Status Monitoring Requirement.....	11
3.2.6 Battery & Power Management Requirement.....	12
3.2.7 Communication Log Requirement	13
3.2.8 Helmet Setup & Calibration Requirement.....	14
3.2.9 Error Detection & Recovery Requirement	15
3.3 Non-Functional Requirements	16
3.3.1 Performance Requirements	16
3.3.2 Security Requirements.....	16
3.3.3 Safety Requirements.....	16
3.3.4 Software Quality Attributes.....	17
3.4 Pseudo Requirements.....	17
3.5.1 Scenarios	18
3.5.1.1 Scenario I.....	18
3.5.1.2 Scenario II	18
3.5.1.3 Scenario III	18
3.5.1.4 Scenario IV	19
3.5.1.5 Scenario V	19
3.5.1.6 Scenario VI.....	19
3.5.1.7 Scenario VII	19

3.5.1.8 Scenario VIII 20

3.5.1.9 Scenario IX..... 21

3.5.1.10 Scenario X 21

3.5.2 Use Case Model.....22

3.5.3 Object and Class Model.....32

3.5.4 Dynamic Models34

3.5.4.1 ARIS System Processing Pipeline..... 34

3.5.4.2 Activity Diagram 35

3.5.5 User Interface - Navigational Paths and Screen Mock-Ups36

4. Glossary.....39

5. References41

1. Introduction

1.1 General Overview

The Artificial Intelligence-Assisted Responsive Interaction System (ARIS) represents a smart communication helmet interface specifically tailored to increase situational awareness, effectiveness, and seamless communication in military or danger scenarios. Handheld communication devices, although widely in use, require manual interaction, which can hinder data processing in critical environments through direct manipulation. In fact, the implemented ARIS combines an intelligent interface that can interpret hand gestures, facial expressions, or voice commands in real time.

At the heart of the ARIS system is an AI-based recognition module that relies on a compact camera and a directional microphone array integrated into the helmet. The camera constantly analyzes visible signs like hand gestures and facial micro-expressions, while the microphone picks up a pre-defined, small vocabulary of voice commands. These are then processed using low-latency machine-learning algorithms that support quick inferences. After recognizing the command, it is transmitted in the form of a secure data packet through an encrypted channel to the central server or reception unit.

Further, there is an online control and monitoring panel in the ARIS. Using such an interface, the authorized root user can monitor the different helmets that are connected as well as check the command logs and monitor the real-time status. Through such an interface, command staff can easily get data from the fields and adapt rapidly to the changing events.

The primary aim of ARIS is to reduce the workload of manual operations, increase the speed of responses, and maintain communication seamlessly in stressful or dangerous environments. It has been developed as a proof-of-concept to show how AI-based communication using multiple media can help in such scenarios where every second counts and traditional communication can cause delays or increase the level of danger.

1.2 Constraints

Economic Constraints

One important aspect of ARIS is that it was developed in an academic or research setting; hence, the utilization of military-grade components in the hardware can easily be ruled out. The use of low-cost components such as Raspberry Pi-based boards, consumer camera modules, and open-

source libraries related to AI helps in developing the prototype but can hinder performance to some extent.

Environmental Constraints

It has been designed to operate in outdoor, military-style environments. It can face challenging lighting conditions, ambient noise, or environmental factors such as dust or abrupt physical shock. Working in extreme temperatures or heavy rain environments is beyond the scope of the current prototype and would need additional ruggedized designs in practical scenarios.

Social Constraints

In the case of ARIS, there are concerns related to privacy issues regarding the collection of biometric data (image and voice recordings). Only authorized personnel have direct access to the data, and all such data needs to be encrypted. Social acceptance can also be obtained through proper communication about the nature and utilization of the system.

Political and Legal Constraints

Military communication frequencies, standards of encryption, and communication channels are closely regulated and are mostly considered classified information. Inevitably, the prototype will function in purely non-classified simulated communication settings. The pertinent regulation the project needs to follow regards wireless communication, encryption, and biometric data protection.

Ethical Constraints

Bias can occur in AI-based recognition technology when trained on imbalanced data sets. To prevent discrimination in outcomes, data sets consulted during training need to include representative skin colors, facial expressions, and hand shapes. Participants in these test procedures need to give valid consent to record audio and video. Surveillance or constant observation without the knowledge of users cannot be allowed.

Health and Safety Constraints

Seeing that the hardware is helmet-based, it has to be lightweight, non-intrusive, and comfortable to wear. The sensors cannot impair the wearer's visibility, nor can the device produce any heat that might cause irritation. All electrical parts of the hardware need to be isolated to eliminate the risk of shocks or electrical shortcuts.

Manufacturability & Technical Constraints

Real-time image and audio processing are challenging tasks for any embedded system like the Raspberry Pi. All hardware components must therefore be easily dis-connectable, maintainable, and replaceable to reduce the number of cycles in model development. In addition, all hardware components must support real-time processing.

Sustainability Constraints

The modular design methodology needs to be adopted in order to easily replace individual components without needing to dispose of the whole system. The use of rechargeable batteries needs to be considered, with the aim of reducing waste, while the design of 3D-printable housings must make use of recyclable materials where possible.

1.3 Professional and Ethical Issues

The issue presented in the ARIS project raises several ethical and professional issues based on the nature of biologic data that the project utilizes.

The protection of privacy for users is a fundamental responsibility. All image and audio data obtained must be anonymized, encrypted, and then deleted after all evaluations are finished. Data obtained in the project can neither be disseminated nor archived beyond the framework of the project. It is the responsibility of the development professionals to follow all ethical requirements, including the Codes of Ethics of both the ACM and the IEEE.

Models trained using small or imbalanced data can result in biased performance. To avoid such issues, the gesture or expression recognition model needs to be tested using diverse data. The limitations of the system need to be identified, and ARIS, in any case, cannot or should not claim complete autonomy or accuracy. All the participants need to be educated regarding the type of data collected, the danger related to it, and the fact that they can withdraw from the process at any stage. The prototype will not be utilized without consent or as any kind of surveillance activity. For the professional developer, there needs to be complete documentation, proper security in development, and all designs need to follow good engineering. ARIS has to be viewed as a prototype stage research tool and in no means reflects something that can or should ever go towards military deployment.

2. Current System

Today, communication in the military relies on hand-carried or worn communication radios. They function satisfactorily, most of the time, but they require manual dexterity in terms of tuning

channels, pushing buttons, or holding microphones, which can always present practical challenges in critical times such as actual combat or in rapid operations.

They lack the feel of the nonverbally communicated messages. Hand gestures, expressions, and other forms of communication can contain vital messages where silence reigns supreme, but conventional radios can't transmit these digitally. They further lack any kind of artificial intelligence; thus, the aspects of multimodal or context-based interactions aren't in play here.

The security level in these devices differs, with many devices not having any kind of logging or centralized management. Currently, commanders lack real-time monitoring facilities where they can check the status of several users at once.

But there remains a gap in the market because there isn't such a solution that integrates gesture detection, facial expression analysis, voice command identification, and secure communication. That was the aim of the ARIS solution.

3. Proposed System

3.1 Overview

The ARIS system is proposed as a wearable, AI-enabled communication solution embedded directly into a helmet. Its purpose is to support soldiers by enabling them to transmit commands without relying on their hands or shifting focus away from their surroundings. The system combines real-time visual recognition, voice command processing, and secure message transmission through a modular architecture.

The system consists of four primary layers:

- **Sensor Layer:** Captures continuous audio and video streams.
- **AI Processing Layer:** Interprets gestures, facial expressions, and voice commands through optimized machine-learning models.
- **Communication Layer:** Encrypts and transmits validated command messages.
- **Interface Layer:** Provides a central web dashboard for monitoring device states, logs, and administrative controls.

Through this layered approach, ARIS aims to reduce reaction time, maintain situational awareness, and provide reliable communication even when the user cannot manually operate traditional equipment.

3.2 Functional Requirements

3.2.1 Visual & Audio Input Capture Requirement

Description:

Through the camera attached on the helmet, the system will continuously record visual input in order to identify hand motions, facial expressions, or pre-programmed visual command patterns. In order to recognize voice commands, the integrated microphone must also record user audio. All captured data must be processed in real-time (1–2 seconds) to ensure uninterrupted communication during field operations.

Trigger:

The helmet is powered on and enters active operation mode.

Preconditions:

- The camera and microphone modules are properly connected.
- The helmet has sufficient battery power.
- The user is within an environment where visual or audio capture is possible.

Postconditions:

- Visual and audio data frames are successfully transmitted to the AI module for processing.

Normal Flow:

1. User wears and activates the helmet.
2. Camera begin streaming visual frames to the processing module.
3. Microphone captures real time audio inputs.
4. System forwards both streams to the AI module within 1–2 second.

Alternative Flows:

- If the camera is obstructed the system notifies the user via auditory alert.
- If noisy conditions are detected the system increases noise filtering sensitivity.
- If either input fail the system switches to the functional input and log the issue.

Non-Functional Requirements:

- **Performance:** Audio or visual capture must have a latency of less than two seconds.
- **Reliability:** Input modules must operate continuously for 8 hours.
- **Accuracy:** Visual or voice data must retain quality under low light or moderate noises.
- **Safety:** The helmet must avoid overheating during the long term input capture.

3.2.2 Command Recognition Requirement

Description:

The system will recognize speech instructions, preset gestures and facial expressions by analyzing the visual and audio inputs using AI based recognition models. Each identified patterns must match a command template found in the system's command database.

Trigger:

AI module receives processed visual or auditory input.

Preconditions:

- Input data successfully captured.
- Command templates exist in the system database.
- The AI recognition module is active.

Postconditions:

- A validated command is generated and forwarded to the communication module.

Normal Flow:

1. System receives visual or audio data from input modules.
2. AI compares the information with voice command and gesture templates that have been stored.
3. A match is found.
4. The system verifies and allocates the proper military command (e.g., advance, halt, request support).
5. The communication module receives the structured command.

Alternative Flows:

- If no match is found the system requests the user to repeat the command.
- If recognition confidence falls below threshold the system log the uncertain input.
- If gesture recognition fails due to lighting fallback to voice command is triggered.

Non-Functional Requirements:

- **Accuracy:** Recognition success must be at least 90%.
- **Performance:** Command validation must occur in under 2 seconds.
- **Security:** Recognition model and templates must be protected from unauthorized modification.
- **Robustness:** The system must be able to function in real time in low light, dust and moving conditions.

3.2.3 Communication Module Requirement

Description:

The helmet and the central command interface will be able to communicate securely in both directions because of the system. Validated commands must be transmitted with a maximum transmission delay five seconds via encrypted wireless channels. Receiving updates or acknowledgments from the base station will also be made possible by the module.

Trigger:

A validated command is produced or an incoming message is received from the control center.

Preconditions:

- User is connected to the central unit via secure wireless link.
- Encryption keys are initialized.
- Helmet communication module is active.

Postconditions:

- Commands are delivered to the control center or received messages are displayed to the user.

Normal Flow:

1. System prepares validated command in structured format.
2. Encrypts the payload using secure algorithm (e.g., AES-256).
3. Sends information to the control center.
4. Gets a response or acknowledgement.
5. Shows the UI's status (delivered, received).

Alternative Flows:

- If wireless signal weakens system retries transmission automatically.
- If communication fully drops system switches to offline logging until reconnection.
- If encryption fails the system halts transmission and warns the user.

Non-Functional Requirements:

- **Performance:** Web interface must respond in 1–2 seconds.
- **Transmission Time:** Commands must reach server in under 5 seconds.
- **Security:** All communication must be encrypted end to end.
- **Availability:** Module must function continuously for 8 hours without overheating.
- **Reliability:** Automatic reconnection must occur within 3 seconds after signal loss.

3.2.4 User Interface Requirement

Description:

Authorized personnel will be able to monitor helmet status, incoming commands, user connectivity and system log in real time through the system's web based control interface. The interface must display connection health, processing delays, live command feeds and unit status within 1–2

seconds. The UI must be minimal, readable, military friendly and optimized for fast situational awareness.

Trigger:

User accesses the central web dashboard.

Preconditions:

- The helmet is connected to the server.
- The operator has valid authentication credentials.
- The web server is online and reachable.

Postconditions:

- The operator can view real time data from all connected helmets.
- UI updates reflect any new command, status change or connection event.

Normal Flow:

1. Operator logs in to the web interface.
2. Dashboard loads helmet lists and their current statuses.
3. Live data (commands, latency, connection time, errors) is displayed.
4. Operator interacts with UI to view detailed logs or system information.

Alternative Flows:

- If a helmet disconnects UI displays “Lost Connection”.
- If server is overloaded UI switches to cached mode with a warning.
- If wrong credentials are entered UI denies access.

Non-Functional Requirements:

- **Performance:** UI responses must render within 1–2 seconds.
- **Usability:** Dashboard must be easy to understand under stress.
- **Security:** Only authorized master users may access logs.
- **Scalability:** UI must support multiple helmets connected simultaneously.

3.2.5 System Status Monitoring Requirement

Description:

The system will track operational metrics about helmets through continuous monitoring which shows processing delays, AI recognition confidence, wireless signal strength and communication delays. The system needs to send an alert to users when its performance metrics fail to meet the set operational thresholds which include latency times exceeding two seconds. Recognition accuracy falls below 90 percent.

Trigger:

New data event is generated from helmet sensors or AI modules.

Preconditions:

- Helmet sensors and module must be active.
- Server must be able to receive telemetry data.

Postconditions:

- System dashboard reflects updated metrics in real time.
- Alerts (visual or audio) are triggered when anomalies occur.

Normal Flow:

1. Helmet collect internal metrics (latency, confidence, connection)
2. Sends telemetry data to server.
3. Server update dashboard instantly.
4. Operator reviews performance.

Alternative Flows:

- If telemetry packet is corrupted system requests retransmission.
- If the system detect overheating risk it warns the operator.
- The system logs reduced performance if AI falls below the accuracy threshold.

Non-Functional Requirements:

- **Accuracy:** Monitoring must reflect real metrics without delay.
- **Reliability:** System must not miss telemetry packets under heavy load.
- **Availability:** Must use small data packets to operate even with constrained internet access.
- **Safety:** Overheating alerts must triggers before damage occurs.

3.2.6 Battery & Power Management Requirement

Description:

In order to guarantee eight hour of continuous operation the system will monitor and control the helmet's power consumption. High consumption alerts, such as a real time battery percentage and estimated remaining time must be present. In order to prolong battery life when necessary, the system must maximize resource utilization (camera, microphone, wireless module)

Trigger:

Helmet is powered on or power related sensor data changes.

Preconditions:

- Battery sensor is functioning.
- Helmet is running in active mode.

Postconditions:

- Updated battery metrics is displayed to the user and the central system.
- Power saving actions may be applied automatically.

Normal Flow:

1. Battery sensor reports current level.
2. System calculates estimated remaining time.
3. Send battery data to UI and server.
4. Operator sees battery status on dashboard.

Alternative Flows:

- If modules consume unusual power system logs abnormal usage.
- If battery sensor fails system displays “Battery Sensor Error”.

Non-Functional Requirements:

- **Performance:** Battery updates must be logged at least every 30 seconds.
- **Reliability:** Power monitoring must not fail during operation.
- **Safety:** System must prevent operation if battery overheats.

3.2.7 Communication Log Requirement

Description:

All incoming and outgoing commands, communication attempts, AI recognition results, latency values and error event will be logged chronologically by the system. The logs will be stored securely so only those authorized can access them from web interface. Commanders should review the logs in order to track field activities and identify issues with either recognition or communication.

Trigger:

A communication event occurs (command validated, command sent, message received, system error)

Preconditions:

- Helmet is connected to the server.
- Secure logging module is active.
- User authentication is completed for viewing logs.

Postconditions:

- A new entry is appended to the communication log.
- Authorized user can review recorded entries in correct time order.

Normal Flow:

1. System generates communication event.
2. Server receives event detail (timestamp, type, helmet ID, latency, status)

3. Log entry is securely stored.
4. Operator opens “Logs” section on web UI.
5. Logs are displayed in sorted timestamped order.

Alternative Flows:

- If server storage is full system delete oldest logs after warning the operator.
- If a log entry fails to save system retries and marks the entry as “pending.”
- If user lacks permission system denies access.

Non-Functional Requirements:

- **Security:** Logs must be encrypted both at rest and in transit.
- **Performance:** Retrieving the log must take less than three seconds.
- **Persistence:** Logs must regularly backed up.
- **Privacy:** Only authenticated operators may view logs.

3.2.8 Helmet Setup & Calibration Requirement

Description:

The system will lead users through an initial calibration process where in users will set up their helmet's camera, microphone, gesture threshold settings, voice sensitivity settings and wireless connection setup. Proper calibration help to provide reliable recognition capabilities and stable communications when used in the field.

Trigger:

User selects “Setup Mode” or turns on the helmet for the first time.

Preconditions:

- Helmet hardware modules must be functional.
- User must follow instruction displayed on the visor or audio guide.

Postconditions:

- Calibration settings is saved locally.
- Helmet transitions automatically to normal operation mode after setup.

Normal Flow:

1. User activates the helmet.
2. System enters setup mode.
3. Calibration begins (camera alignment - microphone test - gesture baseline - voice command baseline network connection test).
4. System stores calibrated values in configuration files.
5. Helmet switches to operational mode.

Alternative Flows:

- If camera calibration fails due to low light system show an alerts the user.
- If network connectivity fail setup pauses until signal is restored.

Non-Functional Requirements:

- **Usability:** Setup should take less than 5 minutes.
- **Reliability:** Calibration values must load correctly every boot.
- **Adaptability:** Calibration must adjust for different environment (military, civilian, indoor, outdoor).
- **Safety:** System must not activate active modules until calibration is complete.

3.2.9 Error Detection & Recovery Requirement

Description:

The system must automatically detect any internal error such as sensor problems or communication dropouts. Some example are sensor failures, communication drops, overheating, corrupted inputs from an AI, delayed processing and battery issues. The system will attempt to correct the detected issue, notify the operator and user and log the incident to facilitate future troubleshooting.

Trigger:

An abnormal condition is detected during operation.

Preconditions:

- Monitoring modules are active.
- Helmet is running in operational mode.

Postconditions:

- User is notified via vibration or audio cue.
- Operator is alerted on the central UI.
- Error is logged and system attempts auto-recovery.

Normal Flow:

1. The system recognizes unusual activity, such as processing longer than two seconds.
2. The error type (communication, sensor, thermal, AI, battery) is categorized by the system.
3. Makes an effort to carry out predetermined recovery actio (retry connection, reset module, reduce load).
4. Logs the error and send alert to central command.
5. Resumes operation if recovery succeeds.

Alternative Flows:

- If recovery fails system enters “Safe Mode.”
- If overheating is detected system shut down camera temporarily to reduce load.

- If wireless fails completely system stores commands locally until reconnection.

Non-Functional Requirements:

- **Reliability:** Recovery actions must execute within 3 seconds.
- **Safety:** System must prevent damage from overheating or battery faults.
- **Availability:** System should remain operational whenever possible via fallback modes.
- **Robustness:** Must function in harsh environments (dust, vibration, low light).

3.3 Non-Functional Requirements

3.3.1 Performance Requirements

The system must operate efficiently under field conditions, supporting continuous visual and audio processing with minimal delays. In order to provide real-time command recognition, the helmet's AI module must process camera and microphone data in 1-2 seconds. Even in unstable network conditions, the wireless transmission of verified orders to the central system cannot take longer than five seconds.

The web interface used by commanders must remain responsive, loading live data, logs, and unit statuses within 1–2 seconds. The system must operate steadily for eight hours straight without overheating or losing AI precision.

Real-time updates (latency values, connection state, recognition confidence) must be reflected instantly on both the helmet and command dashboard.

3.3.2 Security Requirements

The system must incorporate strong security measures to protect sensitive military communication. Before exchanging any data, helmet units and the command server must be needed to authenticate each other. Data transferred between the helmet unit and command server must be securely encrypted (for example AES-256) including command, log, telemetry and real time data.

Only authorized personnel may access the web interface, helmet logs or command history. Role based access control (RBAC) must ensure that master users manage permissions securely. Server configuration, encryption keys and helmet firmware update must be protected against unauthorized access or tampering.

It should also prevent data leakage via secure storage, hashed credential and also encrypted communication channels.

3.3.3 Safety Requirements

The Helmet System has to safely function in severe military environmental condition. It should properly detect and report any overheating, battery malfunction or hardware instability

throughout the 8 hour operation. If abnormal temperature or voltage levels are detected, the helmet must automatically reduce power consumption or temporarily disable high-load modules (e.g., camera) to prevent equipment damage.

To support safe field usage, when serious problems arise, the system must quickly inform the user and the command center so that the user is not dependent on poor communication. Hardware components such as the 1.5–2 kg helmet must meet ergonomic standards to avoid user fatigue or injury.

Real time safety warnings must guide users and rescue personnel to maintain operational continuity and personal safety.

3.3.4 Software Quality Attributes

The system must be robust, fault-tolerant, and highly available. The system must provide clean, user friendly dashboard for operators. Frequent testing, logging and also validation processes must ensure consistent reliability.

The software architecture shall be maintainable and modular. It allowing updates of AI models, recognition template or a change of communication protocol easily without requiring major structural redesign. Security patches and performance update shall be applied regularly to fix emerging vulnerabilities.

3.4 Pseudo Requirements

- The UI should maintain dark and minimal military themed design that help operators quickly understand information during stressful situations.
- Critical warnings and alerts should be highlighted using strong colors such as red so specific events are instantly recognizable.
- The general dashboard should give users experience of stability and reliability therefore enhancing confidence in the performance of the system in the field.
- Error messages should not involve complicated term but should instead lead the user to simple and correct remedial measures.
- Notification sounds and vibration alerts should be easily noticed even in noisy environments. However they mustn't overwhelm the user or break concentration.
- Icons, button and other elements of layout should be sharp, clear and consistent in their look. It feel to provide professional UX experience.
- The feedback given via the helmet should be natural and non intrusive so as not to hinder the user

from moving and acting freely.

- Visual and interaction design need to express key qualities such as reliability, security and operational precisions.

3.5.1 Scenarios

3.5.1.1 Scenario I

Real Time Command Recognition During Field Operation

A soldier in active mission want to give the order like "Move Forward" without using any radio, where voice communication will expose his location.

Action:

Using visual gesture recognition, the ARIS helmet detects the soldier's hand signal in 1-2 second. The AI module validate the gesture and securely sends the command to the command center and connected teammates.

3.5.1.2 Scenario II

Silent Emergency Alert through Facial Expression

Situation:

While reconnoitering the area, soldier detects hostile units in the vicinity and has to alerts the command center without making any noise.

Action:

The soldier makes a pre identified facial expression utilized in silent warnings. The helmet analyzes this visual input in real time. It issues an encrypted distress warning to commanding units, maintaining the stealth of the soldier.

3.5.1.3 Scenario III

Communication Status Confirmation

After sending an earlier message, a soldier wants to check whether the headquarters received the command or not because of network instability in the region.

Action:

The soldier uses voice command as simple as "Check Status." ARIS Helmet processes the audio input and then fetches the most recent message delivery status from the server. Later give the soldier confirmation through vibration.

3.5.1.4 Scenario IV

Low Connectivity Auto-Recovery

Situation:

During mountain operations, the wireless communication signal weakens and hinders temporary data transmission.

Action:

Connectivity drop is automatically detected by the ARIS system, which turns on offline buffering, storing all outgoing messages locally. Once the connection is back the system ll synchronize all data with the command center automatically.

3.5.1.5 Scenario V

AI Recognition in High-Stress Conditions

Situation:

A soldier is ambushed and has to give immediately the order of "Take Cover",while running and under extreme stress.

Action:

The AI of the helmet filter the inputs, despite motion blur and background noise and accurately recognizes the command. It sends high priority alert to teammates and headquarters thus enabling rapid response.

3.5.1.6 Scenario VI

Filtering Incoming Command Logs on the Dashboard

Situation:

The operator of the center wants to review only the "Urgent" or "High Risk" command message sent from various helmets during a chaotic field operation.

Action:

The operator clicks on the filter icon in the web dashboard and selects “High Priority.” The list refreshes and showing only urgent commands. This enables the operator to make fast and informed decisions.

3.5.1.7 Scenario VII

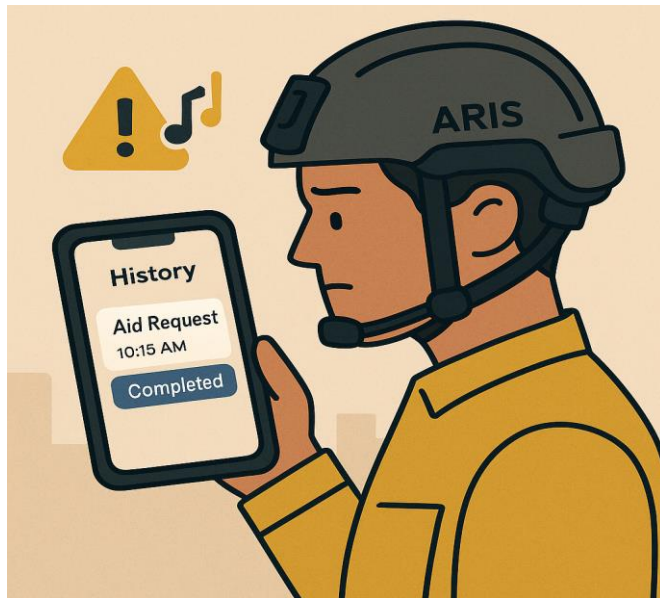
Viewing Previous Helmet Activity History

Situation:

A commander wants to review all action performed by a specific soldier's helmet during the last operation to analyze behavior and communication efficiency.

Action:

The commander opens the "History" tab on the web interface. The system displays a chronological list of past commands, timestamps, processing times and AI recognition outcomes, giving a clear view of the soldier's activities.



3.5.1.8 Scenario VIII

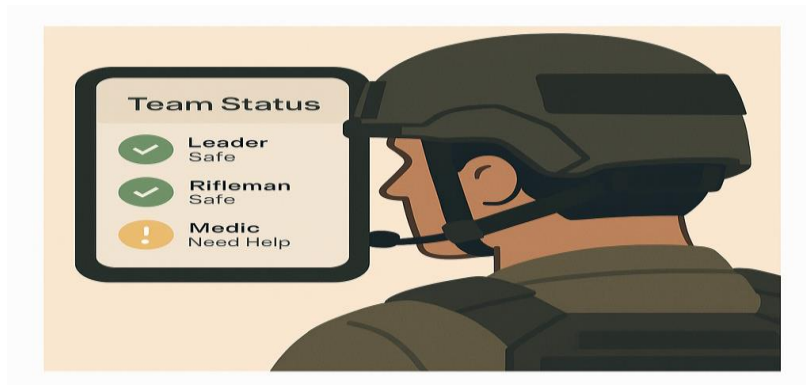
Checking Team Member Status

Situation:

Before starting a coordinated mission, a soldier wants to check if their teammates' helmets are online and connected with the command system.

Action:

The soldier initiates the voice command "Team Status." The ARIS helmet retrieves the most current connectivity data off the server and displays each team member's status as either "Online" "Offline" "Unstable" to inform the soldier on decision making.



3.5.1.9 Scenario IX

Low-Light Facial Expression Recognition

Situation:

During night operations the soldier has to execute “Request Backup” command through a specific facial expression. But the place is poorly lit and so the soldier is anxious about whether the camera can capture the movement clearly.

Action:

The low light optimization mode of the ARIS helmet kicks in and then amplifying the facial feature incoming through the camera. The AI module reads the facial expression without its visibility diminished and send corresponding command to the control center. This ensures that the soldier can still communicate silently in difficult lighting conditions.

3.5.1.10 Scenario X

Handling Low AI Confidence in Command Recognition

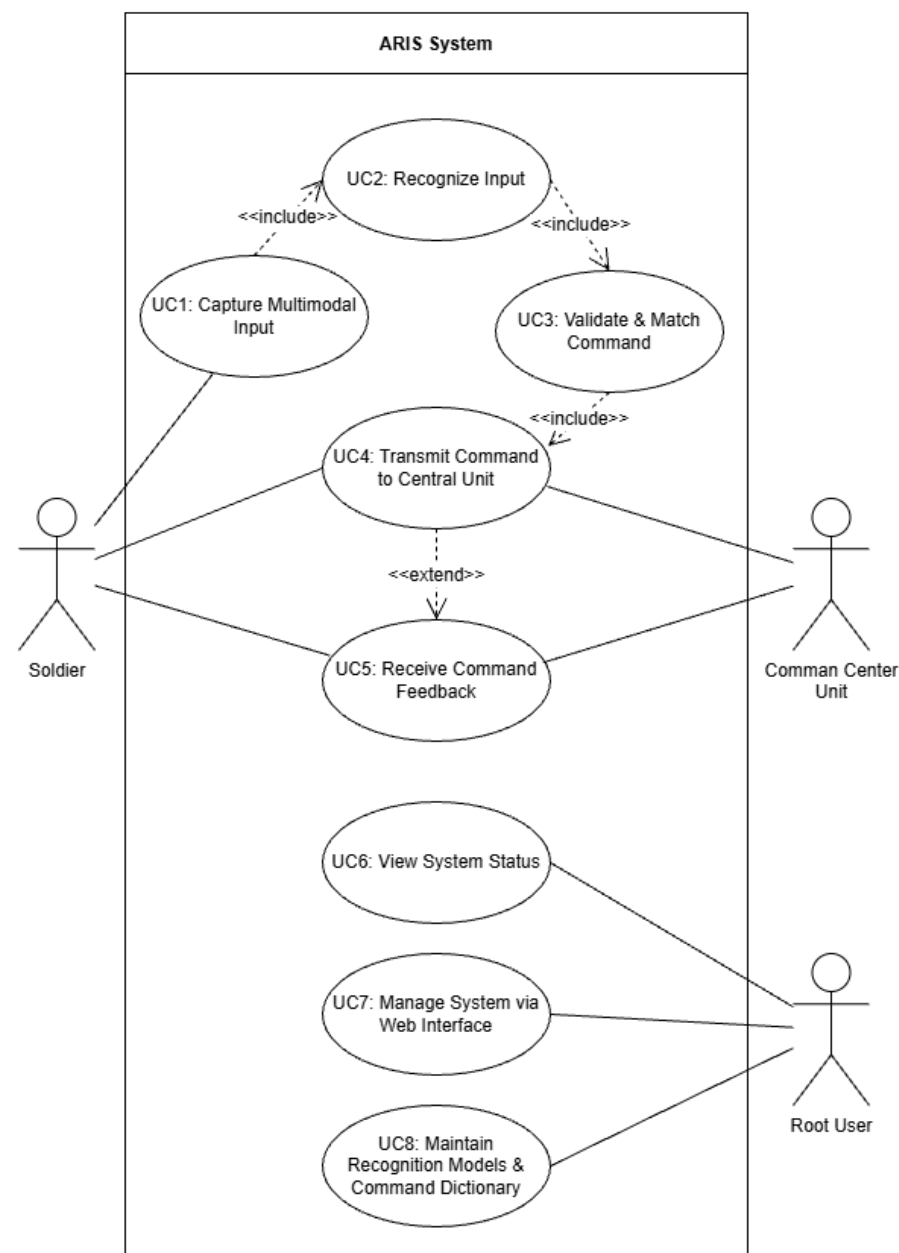
Situation:

A facial expression is made by the soldier which is intended to gives the “Hold Position” command. However, due to rapid movement and partial obstruction of the face the AI model cannot confidently interpret that expression.

Action:

The ARIS helmet detects that the confidence level of recognition is below the acceptable threshold. Instead of sending an erroneous command the system notifie the soldier with short audio cue and requests the repetition of the gesture or expression. In this case, the AI recognizes the command correctly in a second attempts and securely sends it to the command center.

3.5.2 Use Case Model



UC001 Capture Multimodal Input

ID and Name:	UC001 Capture Multimodal Input
Actors:	Soldiers

Description:	The system continuously captures audio and visual input from the soldier via a camera and a microphone placed on the helmet. These unprocessed inputs, which include facial expressions, gestures and voice are collected in real time and forwarded to the recognition pipeline for further processing.
Trigger:	The soldier intends to send a command using face, gesture or voice while the system is running.
Preconditions:	- The helmet is powered on and ARIS software is running. - The camera and microphone are properly connected and functional. - Environmental conditions are within acceptable limits such as moderate lighting and noise level. - Internal modules are initialized and are working properly.
Postconditions:	- Current visual frames and/or audio sections are captured. - The captured input is forwarded to the command recognition module.
Normal Flow:	- Soldier wears the helmet and turns the system on. - System initializes camera and microphone and enters monitoring mode. - System continuously samples video frames from the camera. - System continuously samples audio from the microphone. - When a relevant input window is detected (e.g. the soldier starts speaking or moves in front of the camera), the system collects that part as a multimodal input sample. - The sampled input is packaged and sent to the Recognize Command for analysis.
Alternative Flows:	- Only Visual Input Available: Camera is active but the microphone is muted or not (be able to) used. Therefore, system skips audio capture and only packages visual input. - Only Audio Input Available: Camera input isn't meaningful or missing (e.g. camera angle doesn't show the face and hands). Therefore, system focuses on audio captures. - Low Quality Input: System detects that captured input is too blurred and/or noisy.
Exceptions:	- Camera Failure: Camera cannot be initialized or returns no frames. Only audio interaction continues. - Microphone failure: Microphone cannot be initialized or audio stream fails. - Sensor Overload / Resource Limit: CPU or memory usage exceeds a critical threshold during capture.
Non-Functional Requirements	Usability: System must work while soldier is wearing equipment and shouldn't require manual interaction to start capturing once it is active.

	Performance: Input capture and forwarding must happen with minimal potential so that the total time from user action to command recognition stays within the performance limits.
--	--

UC002 Recognize Input

ID and Name:	UC002 Recognize Input
Actors:	No external actor. (This is an internal system operation.)
Description:	The system receives the raw multimodal input from the Capture Multimodal Input and analyzes it via an AI-based machine learning system. The system first examines and after matches the input with predefined commands from the dictionary. The outcome is a “recognized command candidate”.
Trigger:	UC-001 "Capture Multimodal Input" successfully delivers data.
Preconditions:	- The system is powered on and the recognition module is initialized. - The input from UC001 is valid and ready to process. - The recognition system is working accurately. - The input is clearly categorized and processed. - Predefined command templates exist in the system.
Postconditions:	- The system identifies the type of input. (gesture, facial expression, voice patterns) - A recognized command is produced and forwarded to Validate & Map Command.
Normal Flow:	- The system receives multimodal input captured in UC-001. - The system analyzes the input using its recognition pipeline. - The system determines whether the input matches a known gesture, facial expression, or voice command. - The system generates a recognized command candidate with a confidence score. - The recognized result is sent to the Validate & Map Command.
Alternative Flows:	- Partial Input (audio or visual only): The system runs the appropriate recognition pipeline for the available modality and continues as normal. - Low Confidence Recognition: The system detects the input and matches multiple possible commands. System marks the output as low-confidence and forwards it to UC003.
Exceptions:	Unreadable Input Data: The input cannot be processed (e.g. invalid video frame)

	• Recognition Module Failure: The recognition engine fails to initialize or crashes during processing. • Unsupported Input Pattern: The input doesn't match any known command category.
Non-Functional Requirements	• Performance: Recognition shall occur within the system's processing limit of 2 seconds. • Accuracy: The system shall achieve at least 90% recognition accuracy under normal conditions. • Usability: Recognition must function even when the soldier wears protective gear requiring no manual interaction.

UC003 Validate & Map Command

ID and Name:	UC003 Validate & Map Command
Actors:	No external actor. (This is an internal system operation.)
Description:	The system takes the recognized command candidate produced by UC002 and checks whether it is valid and properly formatted. After validation, system maps the recognized input to a specific predefined command in the command dictionary. Once the command is mapped, it is ready for Transmit Command to Central Unit.
Trigger:	• Successful completion of Recognize Command, which outputs a recognized command candidate.
Preconditions:	• The system has received a recognized command candidate from UC002. • The command dictionary is available and up to date. • Recognition confidence is accessible for validation checks. • The system is fully operational (no errors preventing command validation).
Postconditions:	• A validated and mapped command is produced. • The resulting command is passed to Transmit Command to Central Unit.
Normal Flow:	1. The system receives the recognized command candidate from UC-02. 2. The system checks the recognition of confidence and validity of the input. 3. The system matches the recognized pattern to a predefined command in the command dictionary. 4. The validated command is ready as a structured output. 5. The validated command is forwarded to Transmit Command to Central Unit.

Alternative Flows:	• Low Confidence Candidate: The system sees that the confidence score of recognition is below the boundary. It still attempts to map but tags with “low confidence” and forwards to UC004. • Multiple Possible Matches: The input matches more than one command pattern and selects the highest-confidence match.
Exceptions:	• No Matching Command Found: The system cannot map the recognized pattern to any command in the dictionary. It outputs an “unmapped command” error. • Corrupted Candidate Data: The command candidate from UC002 is incomplete. The system discards the sample. • Command Dictionary Failure: The command dictionary cannot be accessed (e.g. missing files). The system terminates validation and reports an error.
Non-Functional Requirements	• Accuracy: Mapping must be deterministic and consistent for matching recognized inputs. • Security: Only authorized commands may be validated. Invalid or external data must not be accepted. • Maintainability: The mapping rules and command dictionary must be easily updatable.

UC004 Transmit Command to Central Unit

ID and Name:	UC004 Transmit Command to Central Unit
Actors:	Soldiers
Description:	This use case handles the secure transmission of the validated command to the central unit. System encapsulates the command into a structured format and sends it after encryption through a wireless communication channel. The central server receives commands for further processing.
Trigger:	• Successful completion of UC003 Validate & Map Command which is producing a ready-to-send command.
Preconditions:	• A validated command exists in the system. • Wireless communication channel is active and operational. • Soldier’s device and the central server are within communication range of each other. • Mutual authentication between device and server has been completed.
Postconditions:	• The command is encrypted and transmitted to the central unit. • The central unit receives and processes the command.

	• A success event may trigger Receive Feedback.
Normal Flow:	1. The system receives a validated command from UC003. 2. The system prepares a structured message. 3. The system encrypts the message using the configured security algorithm. 4. The device establishes a secure connection with the central unit. 5. The encrypted command is transmitted wirelessly. 6. The central unit receives the message. 7. The system prepares the transmission result. 8. The system may wait for a response that triggers UC005.
Alternative Flows:	• Retransmission Due to Weak Connection: Transmission fails due to low signal quality. The system retries sending the same command. • Offline Mode (Temporary Disconnection): The central server is unavailable. The system queues the command temporarily. The command is transmitted automatically once connection is restored.
Exceptions:	• Encryption Failure: Encryption module fails so the system cannot transmit. System logs the error and cancels the operation. • Transmission Interrupted: Connection breaks mid-transmission. • System retries or cancels the process.
Non-Functional Requirements	• Reliability: System must auto-retry or recover from minor communication errors. • Privacy: No unauthorized device may intercept or read the transmitted data.

UC005 Receive Feedback/Command

ID and Name:	UC005 Receive Feedback/Command
Actors:	Soldiers
Description:	This use case describes the process where the soldier's device receives feedback or commands sent from the central unit. The system continuously listens to encrypted messages and decrypts, understands the content, and presents the result to the soldier (e.g. confirmation, warning or new instruction).
Trigger:	• Central server sends feedback or commands after UC004 transmission. • Server initiates communication (e.g. sends alert or command)
Preconditions:	• A secure communication channel is established.

	• The helmet system is powered on and actively listens to incoming messages. • Encryption/decryption modules are functional. • The central server is online and able to send messages.
Postconditions:	• Incoming data is successfully received and decrypted. • Feedback or command is decoded by the system. • The appropriate output is transmitted to the soldier.
Normal Flow:	1. The system monitors the wireless channel for incoming data. 2. The central unit sends feedback or command. 3. The device receives an encrypted message. 4. The system decrypts the message and verifies its integrity. 5. The message content is analyzed (e.g. new instruction, alert). 6. The feedback is shown to the soldier through the appropriate interface (sound).
Alternative Flows:	• Delayed Feedback: The message arrives later than expected due to network delay. System processes it normally once received. • Multiple Feedback Messages: The server sends multiple updates quickly. The system queues and processes them in order.
Exceptions:	• Decryption Failure: The message cannot be decrypted due to a key mismatch. System logs the error. • Connection Loss: Connection drops before or during reception. System retries or waits until connection is restored. • Invalid Message Format: Received data does not match expected packet structure. System ignores the message.
Non-Functional Requirements	• Security: All messages received must be encrypted and authenticated. • Performance: Feedback should be processed with minimal delay to maintain real-time responsiveness.

UC006 View System Status

ID and Name:	UC006 View System Status
Actors:	Root User (Admin)
Description:	This use case describes how the root user views the overall status of the ARIS system through the web interface. The system presents current connection states, active devices and other relevant monitoring information. The purpose is to allow the admin to observe condition of the system and ensure stable operation.

Trigger:	The root user logs into the web interface and selects the system status or dashboard section.
Preconditions:	- The root user is authenticated. - The web interface is running and connected to the central server. - Monitoring modules (status tracking, logs, connection manager) are active.
Postconditions:	- System status information retrieved and displayed to the admin. - The root user gets statistics about the system's current health and connection activity.
Normal Flow:	- The root user opens the web interface. - The user navigates to the dashboard page. - The system retrieves data such as connection state, uptime and active devices. - The system displays the information in a readable format (tables, icons, color codes, etc.). - The root user reviews the system status.
Alternative Flows:	- Partial Status Available: Some modules respond, but others fail. The system displays available information. - No Active Devices Connected: No helmets or users are currently connected. The dashboard displays a "No active connections" message.
Exceptions:	- Server Unreachable: Web interface cannot access the central server. The system displays an error message. - Data Retrieval Error: Monitoring services fail to return data. The system logs the error.
Non-Functional Requirements	- Reliability: Status information must remain consistent even with multiple user connections. - Usability: System status information must be shown in a clear, easy-to-understand format using clear visual signals. - Maintainability: Status displays shall update automatically whenever new system modules are added or existing ones are changed.

UC007 Manage System

ID and Name:	UC007 Manage System
Actors:	Root User (Admin)

Description:	This use case describes how the root user manages the system through the web interface. The admin can monitor active connections, manage permissions, configure system settings, and perform general organizational tasks. The goal is to ensure stable operation and proper control over the ARIS system.
Trigger:	The root user logs into the web interface and selects the management or configuration panel.
Preconditions:	- The root user is authenticated. - The web interface and server backend are operational. - Necessary configuration files or modules are loaded. - Network connection between client and server is active.
Postconditions:	- Requested configuration or management actions are successfully applied. - System state is updated accordingly.
Normal Flow:	- Root user logs into the web interface. - User navigates to the system management section. - The system displays available management options (connections, permissions, settings). - Root user selects an action (e.g. adjust configuration, manage devices, update permissions). - System applies the selected change. - The result is shown to the user.
Alternative Flows:	- User Attempts Unauthorized Action: The root user tries to make a change or access a feature that their permission level does not allow. System blocks the action and displays an authorization warning. - Partial Update: Only part of a configuration change succeeds. System displays a warning and suggests corrective actions.
Exceptions:	- Server Timeout: Server does not respond. The system informs the user about the timeout and safely cancels any unfinished or pending updates to prevent incorrect configurations. - Configuration File Error: The system cannot properly read or update the required configuration files. It alerts the user that the configuration process cannot continue - Web Interface Crash: The web interface crashes during an operation. - The system stops the ongoing action, informs the user about the issue.
Non-Functional Requirements	Usability: The interface must be simple and intuitive so that settings can be managed easily.

	• Maintainability: Management functions should support future system additions without major changes. • Security: Only authenticated and authorized users may change system settings.
--	--

UC008 Maintain Recognition Models & Command Dictionary

ID and Name:	UC008 Maintain Recognition Models & Command Dictionary
Actors:	Root User (Admin)
Description:	This use case explains how root user maintains and updates the system's settings about recognition process, such as gesture/face/voice recognition parameters, and edits or expands the command dictionary. The purpose is to improve recognition accuracy, add new commands, or update existing ones without modifying the main system.
Trigger:	• The root user selects the "Recognition Settings" or "Command Dictionary" section in the web interface.
Preconditions:	• The root user is authenticated and authorized to perform system maintenance tasks. • Server and web interface modules are operational. • Configuration files are accessible and not corrupted.
Postconditions:	• Updated recognition settings or command dictionary changes are saved. • The system loads the new configuration and applies it for future recognition tasks.
Normal Flow:	1. Root user accesses the maintenance section via the web interface. 2. The system displays current recognition settings and command dictionary. 3. Root user selects an item to update or add new entries. 4. System validates the changes (e.g. correct format). 5. Updated settings or dictionary entries are saved to configuration files. 6. The system reloads the updated configuration. 7. User receives confirmation that the updates were successful.
Alternative Flows:	• Minor Validation Issues: Some entries contain warnings (e.g. duplicate naming). System highlights these issues and requests correction before saving.
Exceptions:	• Write/Save Failure: The system cannot save the updated configuration files. It notifies the user and keeps the previous version.

	• Invalid Configuration Format: The updated settings contain syntax or formatting errors. The system rejects the changes.
Non-Functional Requirements	• Maintainability: Updates should be easy to apply without modifying the system core. • Reliability: Changes must be saved consistently without corrupting existing settings. • Scalability: The system should support adding new commands or recognition models without redesign.

3.5.3 Object and Class Model

ARIS System Objects

- **Soldier (Helmet User):** A field user who creates commands through gestures, facial expressions or voice inputs.
- **ARIS Helmet:** A unit that is capable of AI processing and has imaging, audio, sensing, and wireless components.
- **Root User (Administrator):** The person who has the right to use the system.
- **Command Message:** The data packet that is formed in a specific way when a gesture, expression, or voice input is recognized and mapped to a command already defined.
- **Recognition Input:** The visual and audio information that is captured by the helmet's camera and microphone during the recognition process.
- **Communication Channel:** The encrypted wireless link that is shown between ARIS Helmet and Central Command Unit, and is used for two-way communication.
- **System Session:** The active connection that keeps track of authentication status, link quality, and general session parameters during communication.
- **Log Entry:** A record that is timestamped and contains command transmissions, errors, delays or telemetry metrics. It is used in UC005, UC006, and UC007.

Classes in the ARIS System

- **InputManager:** Gathers the unprocessed visual and audio data from the helmet's sensors, operates continuously to look for the input.
Use Case: UC001 – Capture Multimodal Input.
- **VisionProcessor:** Processes the visual data to recognize gestures and detect facial expressions.
Use Case: UC002 – Recognize Input.
- **AudioProcessor:** Treats the audio signals and retrieves voice command segments.
Use Case: UC002 – Recognize Input.

- **RecognitionEngine:** Merges the visual and the audio outputs to produce one acknowledged command candidate together with the corresponding confidence score.
Use Case: UC002 – Recognize Input.
- **CommandValidator:** Checks if the recognized candidate has the necessary confidence and structure criteria.
Use Case: UC003
- **CommandMapper:** Aligns the input that has been validated with the relevant predefined command in the command dictionary and creates the final structured command message.
Use Case: UC003
- **CommandDatabase:** Keeps all predefined templates for gestures, facial expressions, and voice commands stored.
Use Cases: UC003 and UC008.
- **CommunicationModule:** Commands that go out are encrypted, transmitted through the wireless channel, and incoming messages from the central unit are decoded.
Use Cases: UC004 and UC005.
- **EncryptionService:** The service employs advanced cryptographic algorithms like AES-256 for the protection of data during transmission.
Use Case: UC004.
- **FeedbackHandler:** In the process, the central unit sends encrypted feedback or instructions and the user receives the interpreted response after decryption and the whole process is done by the handler.
Use Case: UC005.
- **StatusMonitor:** The monitor is capable of measuring the performance metrics such as latency, signal quality, recognition confidence and temperature.
Use Case: UC006.
- **SessionManager:** The manager is responsible for creating sessions, authenticating users, recovering links, and managing re-connections.
Use Cases: UC004, UC005 and partly UC006.
- **UserManager:** The user authentication, role assignment, and permission control for root user.
Use Cases: UC006, UC007 and UC008.
- **WebInterfaceController:** Controls dashboard functions including status monitoring, log display, system management and model updates.
Use Cases: UC006, UC007 and UC008.
- **LogManager:** Responsible for organizing logs of commands, errors, delays, and system events with timestamps.
Use Cases: UC005, UC006, and UC007.

- **ConfigManager:** Stores and updates configuration files, the command dictionary and recognition model settings.
Use Case: UC008 – Maintain Recognition Models & Command Dictionary.

3.5.4 Dynamic Models

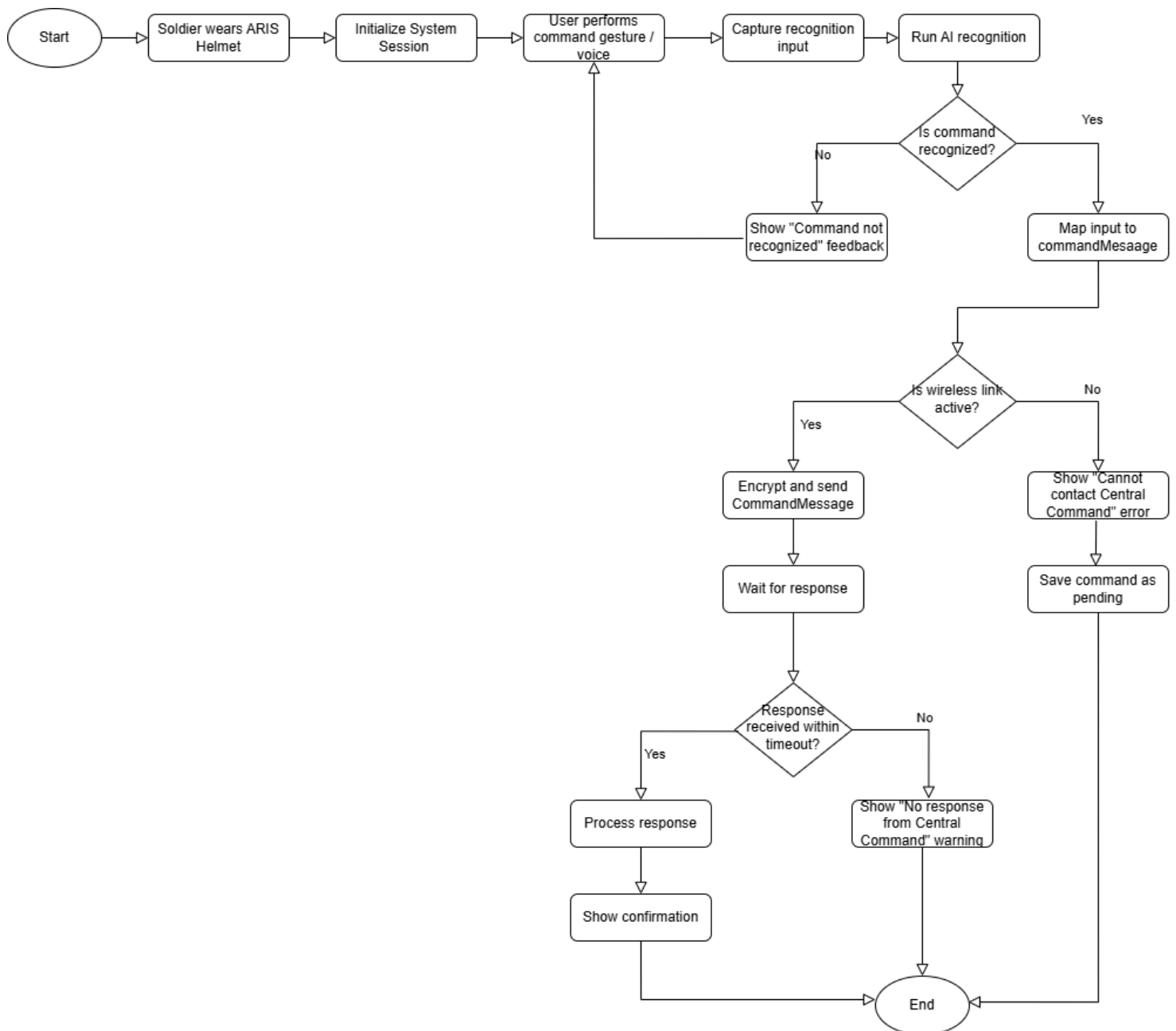
3.5.4.1 ARIS System Processing Pipeline

ARIS is a system that processes multimodal inputs from soldiers during their operations in real time through a processing pipeline. When an event of input is recognized, the system starts a predefined series of actions for data analysis, command generation, and secure sending to the central unit:

1. **Input Capture:** The camera and microphone of the helmet are engaged in constant monitoring. The system will take out the corresponding multimodal window for processing as soon as it detects a gesture or a vocal segment that is of interest.
2. **Pre-Processing:** The visual data is brought to a common level, made more visible under low-light conditions, and is then filtered for clarity. The audio is the subject of noise reduction and its fragmentation in order to obtain the part that contains the command.
3. **Recognition:** The visual and audio processors' results are fed into the RecognitionEngine, which categorizes the input as either a gesture, facial expression or voice command. A command candidate along with its trust score is given.
4. **Validation and Mapping:** The CommandValidator checks whether the candidate meets the required confidence threshold. In case of approval, the CommandMapper does two things; first identifies the input with the corresponding predefined command and then generates a structured command message.
5. **Secure Transmission:** The CommunicationModule makes use of AES-256 encryption for the command and sends it off to the central command unit. In the event of poor connectivity, either the system will keep on trying to send the message or it will hold the command temporarily and wait for the link to be restored before sending it through.
6. **Feedback Processing:** The main control unit might return a confirmation, a warning, or an order. The FeedbackHandler is responsible for handling the incoming communication by first decoding and then forwarding it to the user through the selected notification channel.
7. **Logging and Monitoring:** The system records commands and telemetry values; all these are documented. The logs are made available to the top user through the online dashboard for operational awareness. The pipeline covers the areas of real-time detection, secure communication, and monitoring as specified in the respective Use Cases.

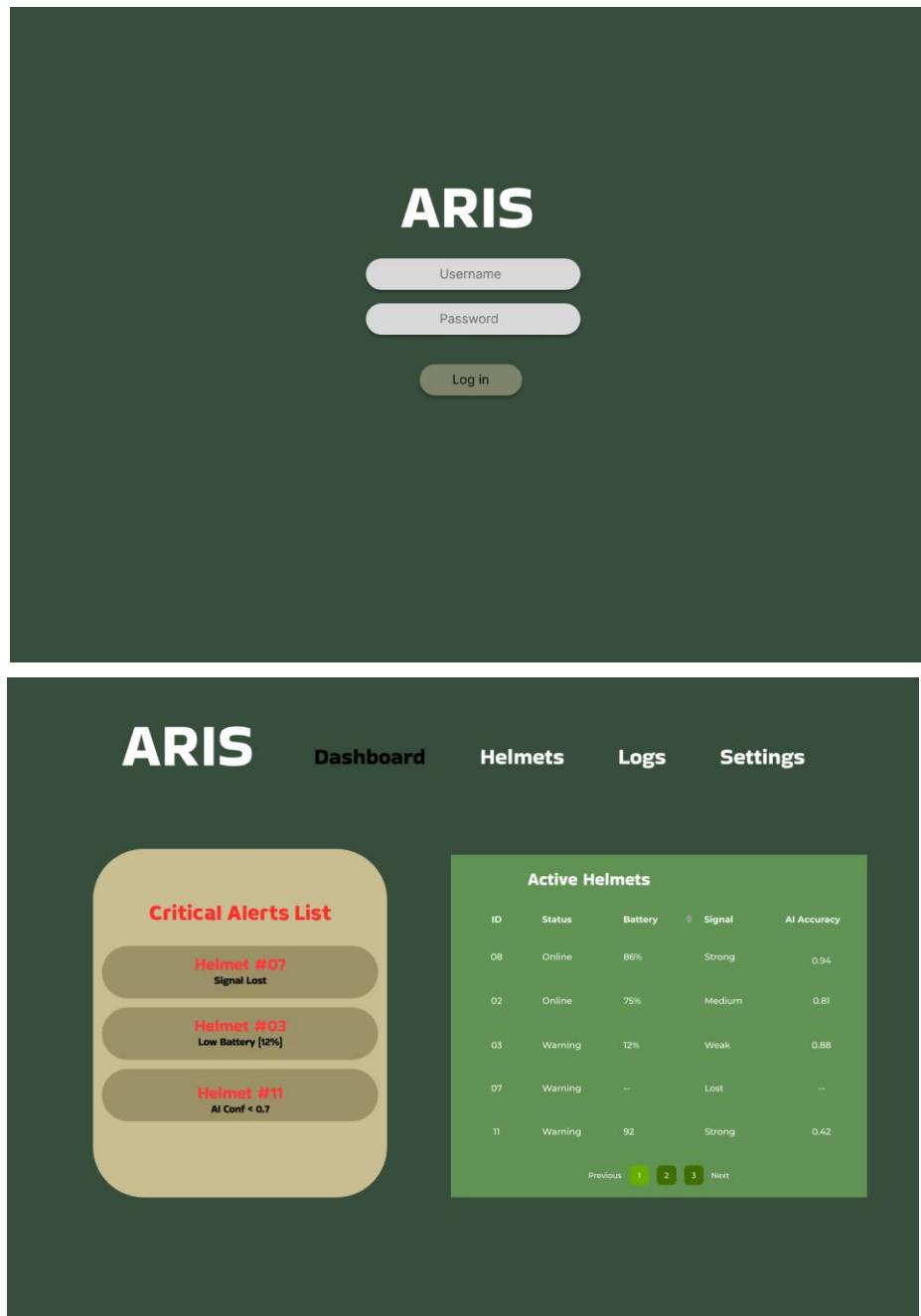
3.5.4.2 Activity Diagram

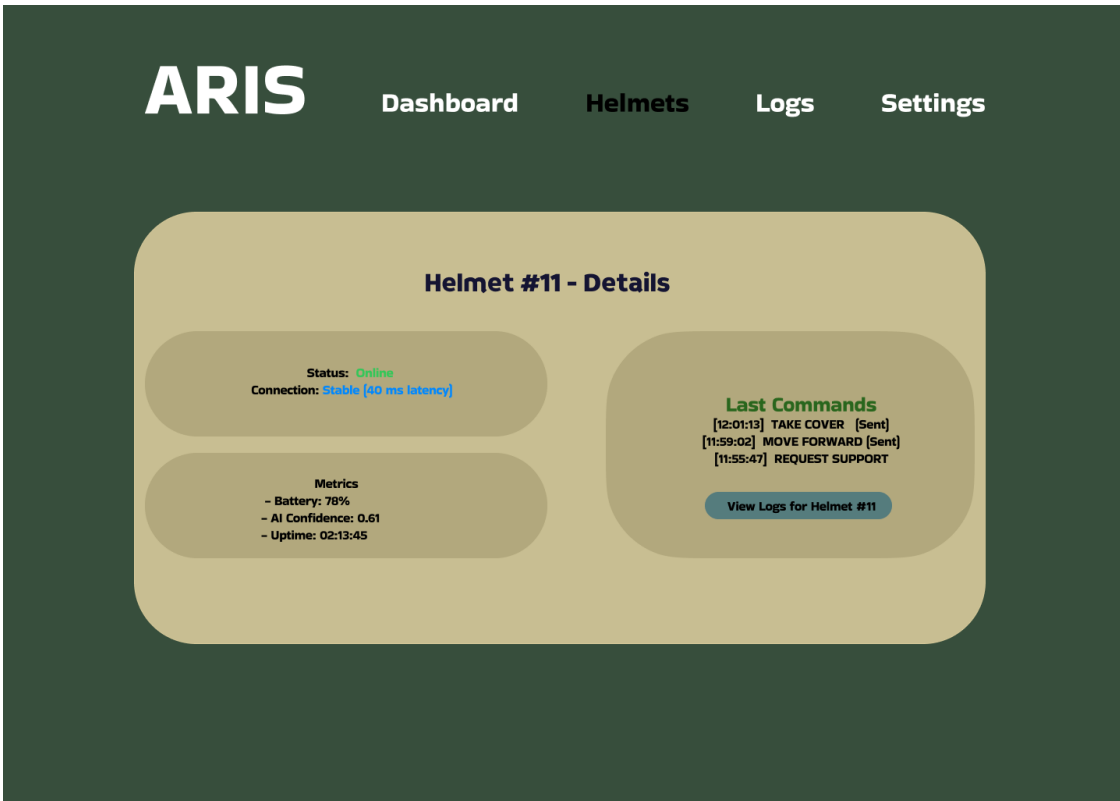
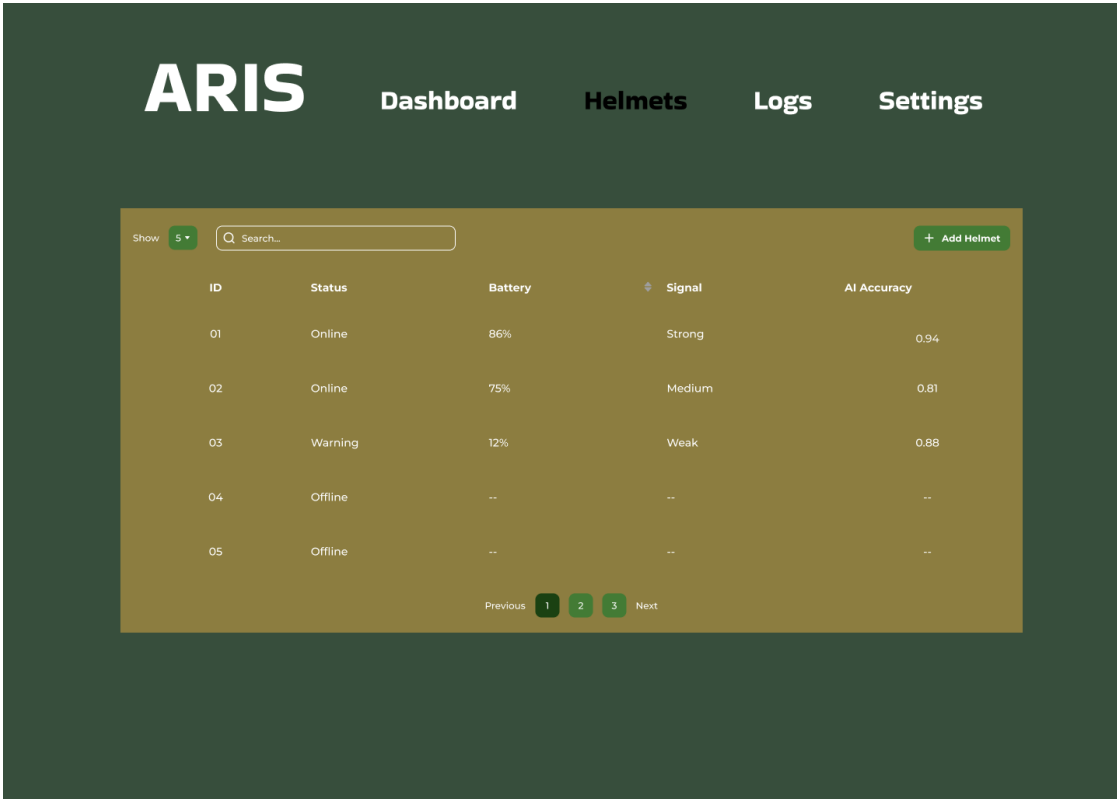
The diagram of activities depicts the way the ARIS Helmet handles an AI-based order. After the fighter gives a sign or call command, the machine goes through the input by the use of AI recognition. In case the command is not acknowledged, the helmet provides feedback and requests the user to say it again. If it is acknowledged, the system produces a CommandMessage and verifies network connection. Offline commands receive an error or are added to the list of pending ones, while the online ones are encrypted and dispatched to Central Command. The helmet then awaits a reply, shows the user the outcome, and concludes the operation.

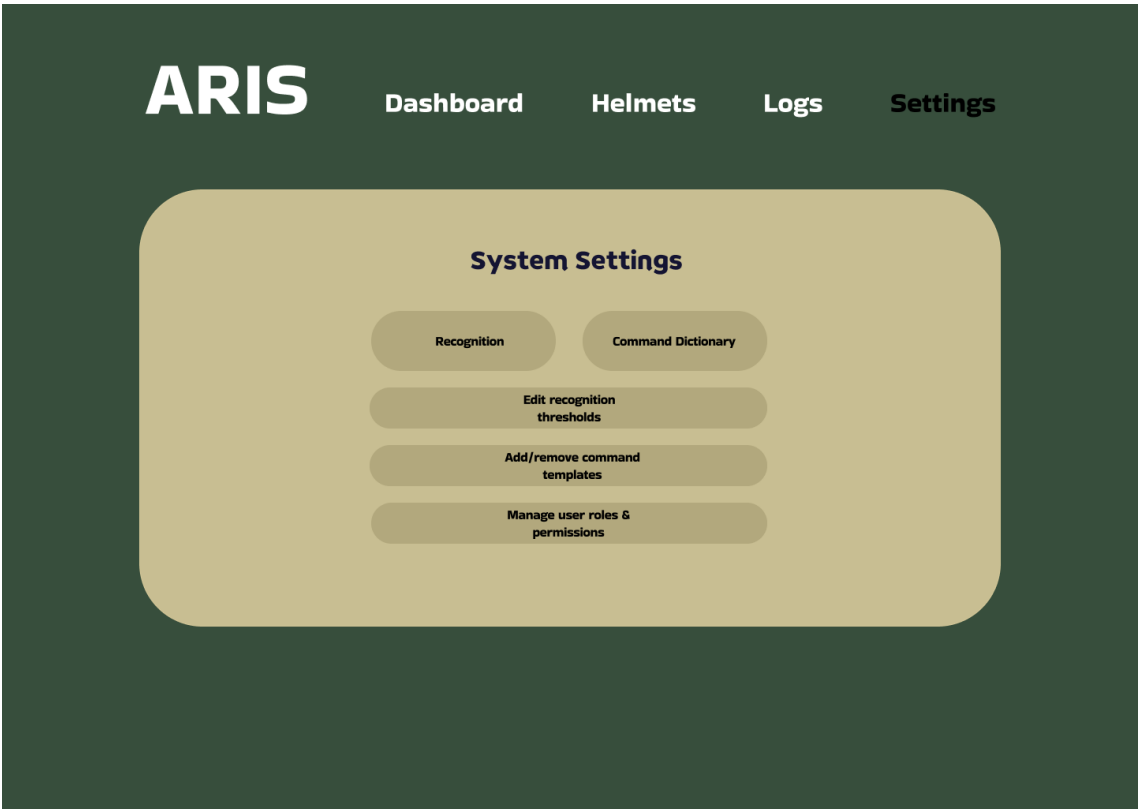
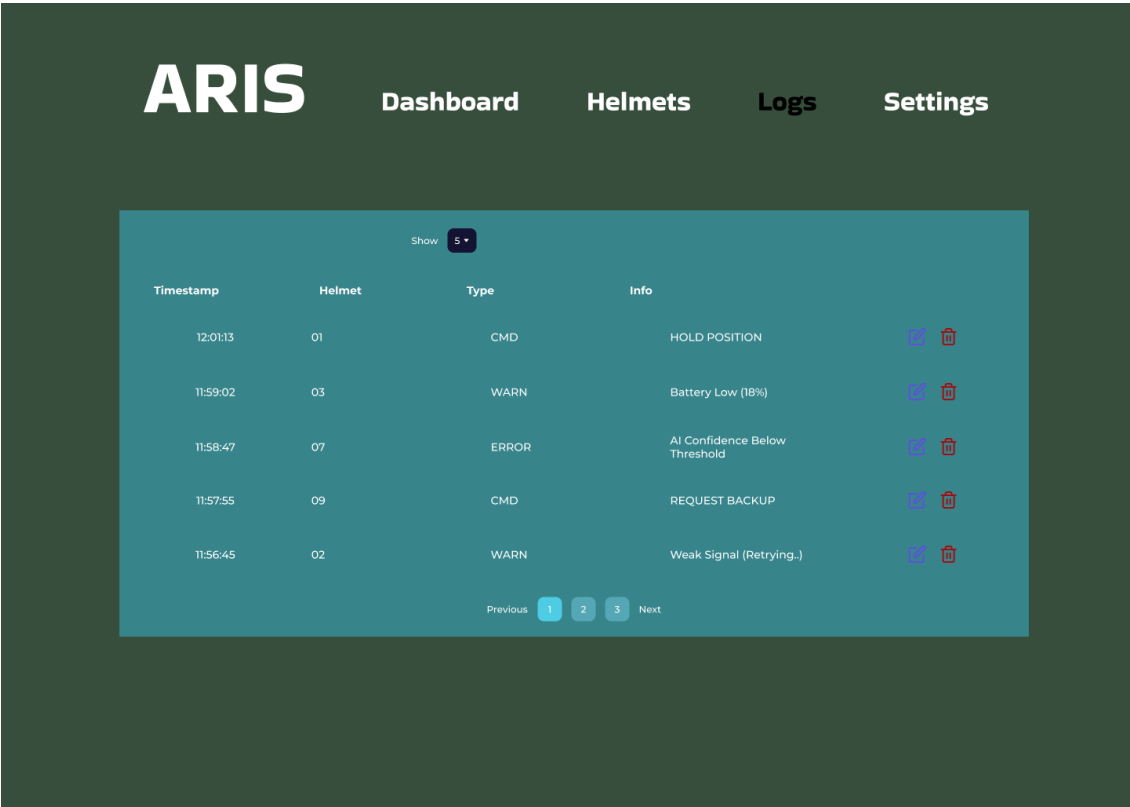


3.5.5 User Interface - Navigational Paths and Screen Mock-Ups

The presented mock-up screens display the suggested user interface design for the ARIS web-based control system. Every screen encapsulates the functional and non-functional requirements that have been set in the specification, such as dark military-themed visuals, real-time status visibility, secure user access, and distinct critical alerts. The navigation and layout are designed to maximize operator awareness while reducing cognitive load in high-stress field operations.







4. Glossary

ARIS Helmet: An advanced digital helmet that soldiers wear in the field and comes with a camera, a microphone, communication hardware and AI processing ability.

Multimodal Input: All the inputs that are taken through different sensing channels such as hand movements, facial expressions and spoken words.

Visual Input: Image frames captured by the helmet-mounted camera, used for gesture and facial expression recognition.

Audio Input: The sound that the microphone captures and is used for detecting and understanding voice commands is called sound data.

Recognition Pipeline: The chain of AI-driven actions involved in analyzing the visual and audio inputs, feature extraction, and classification of the features into specified commands.

Gesture Recognition: A camera captures particular hand movements and the corresponding preset commands are attributed by a process of recognition.

Facial Expression Recognition: The movements of a soldier's face are studied to reveal signal patterns of silent communication that are on the list of the predefined ones.

Voice Command Recognition: The spoken commands that are on the list of predefined ones are recognized and classified by means of processing the audio signal.

Command Candidate: The first result of the recognition pipeline that shows the command with a potential match together with the confidence score.

Confidence Score: The number that tells the degree of AI model's certainty that the recognized input is identical to the command in the predefined set.

Validation: A confirmation process of whether the recognized command candidate meets required thresholds and structural rules before being accepted.

Command Mapping: The matching of a validated input to a predefined military command stored in the system's command dictionary is what this task is all about.

Command Dictionary: It is a structured repository of all predefined gestures, facial expressions, and voice commands that can be recognized and executed by the system.

Command Message: The mapped structured command packet, ready to be encrypted and transmitted to the central unit, is thus generated.

Encryption: The conversion of command messages into a secure encoded form to protect them from unauthorized access during transmission.

Communication Module: The part that handles the encrypting, sending and receiving of commands via a secure wireless channel between the helmet and the central command unit.

Central Command Unit: The primary control system that is responsible for receiving commands from helmets, processing them and, if required, returning feedback or operational instructions.

Feedback Message: Confirmation, alert or instruction transmitted by the central unit to the soldier's helmet via the secure communication channel.

Telemetry Data: Operational metrics that are transmitted from the helmet to the server, including the values for latency, signal strength, AI confidence, and internal sensor readings.

Root User (Administrator): The designated system operator with the right to see the system status, regulate the access, support the recognition models, and examine the communication logs.

Session: A secure communication channel set up between the helmet and the central command unit continued during the active operation.

Latency: The time lag measured between the capturing of input, recognition, transmission and the system's response.

Error Recovery: The automatic procedures that the system follows to restore operation after it has been interrupted by sensor malfunction, connection loss, or overheating.

Log Entry: A unit of information that has been recorded and contains the details of a command, system status update, warning or error for auditing and analytics.

Offline Mode: A backup state where the helmet saves commands temporarily in local memory when wireless connectivity is not available and sends them once the connection is back.

Calibration: The initial setting phase in which the camera, microphone, gesture baselines, and voice sensitivity settings are configured before the regular operation begins.

5. References

Raspberry Pi Foundation, “Raspberry Pi 4 Model B Datasheet,” Raspberry Pi, 2023. [Online]. Available: <https://www.raspberrypi.com/documentation/>

Institute of Electrical and Electronics Engineers, “IEEE Code of Ethics,” IEEE, 2020.

Object-Oriented Software Engineering, Using UML, Patterns, and Java, 2nd Edition, by Bernd Bruegge and Allen H. Dutoit, Prentice-Hall, 2004, ISBN: 0-13-047110-0.

Week 6 – System Modeling & UML Diagram Types, SENG214 Course Notes, Department of Computer Engineering, TED University, 2025.